



Teoler High School

ADVANCED TEST-4

Session 2026 - 27

Grade - X - B

Time : 1 Hours

Maximum Marks : 60

Instructions :

Date : 24-08-2026

SECTION : I

[MULTIPLE CORRECT TYPE]

Mathematics : MCQ(Multiple correct answer) - 03 questions

Physics : MCQ(Multiple correct answer) - 03 questions

Chemistry : MCQ(Multiple correct answer) - 03 questions

This section contains THREE (03) questions. Each question has FOUR options (A), (B), (C) and (D). ONE OR MORE THAN ONE of these four option(s) is(are) correct answer(s). For each question, choose the option(s) corresponding to (all) the correct answer(s). Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 ONLY if (all) the correct option(s) is(are) chosen;

Partial Marks : +3 If all the four options are correct but ONLY three options are chosen;

Partial Marks : +2 If three or more options are correct but ONLY two options are chosen, both of which are correct;

Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);

Negative Marks : -2 In all other cases.

SECTION : II

[NON-NEGATIVE INTEGER TYPE QUESTIONS]

Mathematics : - 02 questions

Physics : - 02 questions

Chemistry : - 02 questions

This section contains TWO (02) questions. The answer to each question is a NON-NEGATIVE INTEGER. For each question, enter the correct integer corresponding to the answer. In case your answer is in decimal then round off your answer to the nearest integer. Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct integer is entered;

Zero Marks : 0 In all other cases.

(MATHEMATICS)
SECTION - I
[MULTIPLE CORRECT CHOICE TYPE]

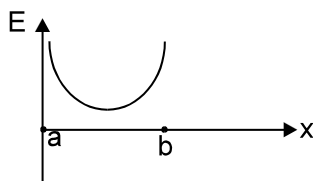
1. The values of k for which the equation $x^2 + 2(k-1)x + k + 5 = 0$ has atleast one positive root, are:
 (A) $[4, \infty)$ (B) $(-\infty, -1] \cup [4, \infty)$ (C) $[-4, -1]$ (D) $(-\infty, -1]$
2. Let $S_n (n \geq 1)$ be a sequence of sets defined by $S_1 = \{0\}$, $S_2 = \left\{\frac{3}{2}, \frac{5}{2}\right\}$, $S_3 = \left\{\frac{8}{3}, \frac{11}{3}, \frac{14}{3}\right\}$,
 $S_4 = \left\{\frac{15}{4}, \frac{19}{4}, \frac{23}{4}, \frac{27}{4}\right\}$ then
 (A) third element of S_{20} is $\frac{439}{20}$ (B) third element of S_{20} is $\frac{431}{20}$
 (C) sum of the elements in S_{20} is 589 (D) sum of the elements in S_{20} is 609
3. If $\sum_{r=1}^n r(r+1)(2r+3) = an^4 + bn^3 + cn^2 + dn + e$, then
 (A) $a + c = b + d$ (B) $e = 0$
 (C) $a, b - \frac{2}{3}, c - 1$ are in A.P (D) c/a is an integer

SECTION - 2
[NON-NEGATIVE INTEGER TYPE].

4. The least positive integer satisfying the equation $|2x| - |x - 4| = x + 4$, is
5. If $2x = y^{1/5} + y^{-1/5}$ and differential equation corresponding to given equation is
 $(x^2 - 1) \frac{d^2 y}{dx^2} + x \cdot \frac{dy}{dx} = ky$, then find the value of k .

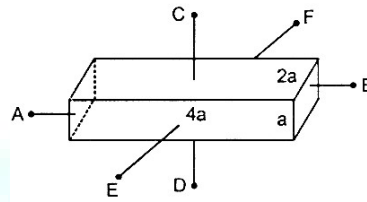
(PHYSICS)
SECTION - I
[MULTIPLE CORRECT CHOICE TYPE]

6. Two point charges a & b , whose magnitudes are same are positioned at a certain distance from each other with a at origin. Graph is drawn between electric field strength at points between a & b and distance x from a . E is taken positive if it is along the line joining from a to b . From the graph, mark the incorrect options:



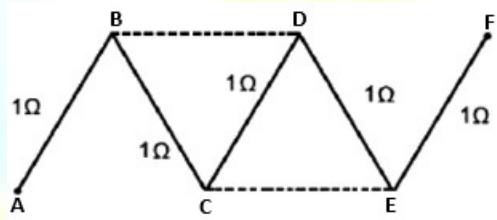
- (A) a is positive, b is negative (B) a and b both are positive
 (C) a and b both are negative (D) a is negative, b is positive

7. A conductor having conductivity σ with rectangular cross section has dimension $(4a \times 2a \times a)$ as shown in figure. Then,



- (A) Resistance across EF is $\frac{1}{2\sigma a}$. (B) Resistance across AB is $\frac{2}{\sigma a}$.
 (C) Resistance across CD is $\frac{1}{8\sigma a}$.
 (D) Resistance across any two opposite ends will be same for same conductor.

8. A circuit consists of five identical conductors ($R = 1\ \Omega$) as shown in figure. Then, two similar conductors are added as indicated by dotted lines (between B to D and C to E). If the ends A and F are connected by a battery of 10 volts

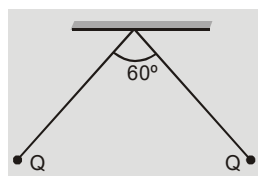


- (A) The ratio of resistances before and after addition will be $\frac{5}{3}\ \Omega$
 (B) Current through CD will remain same before and after addition.
 (C) Current through BC before addition will be 2A.
 (D) Current through BD will be $\frac{5}{3}$ A after addition.

SECTION - 2

[NON-NEGATIVE INTEGER TYPE].

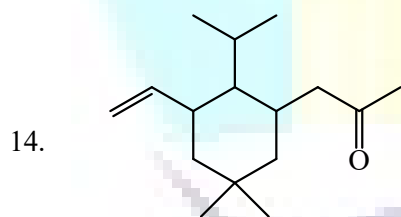
9. A charged water drop of radius $0.1\ \mu\text{m}$ is under equilibrium in some electric field. The charge on the drop is equivalent to electronic charge. The intensity of electric field (in N/C) is ($g = 10\ \text{m/s}^2$) (Nearest integer)
10. Two small spherical balls each carrying a charge $Q = 10\ \mu\text{C}$ are suspended by two insulating threads of equal lengths 1m each, from a point fixed in the ceiling. If it is found that in equilibrium the threads are separated by an angle 60° between them, as shown in the fig. The tension in the threads is given by $\frac{9x}{10}$, where x is:



(CHEMISTRY)
SECTION - I
[MULTIPLE CORRECT CHOICE TYPE]

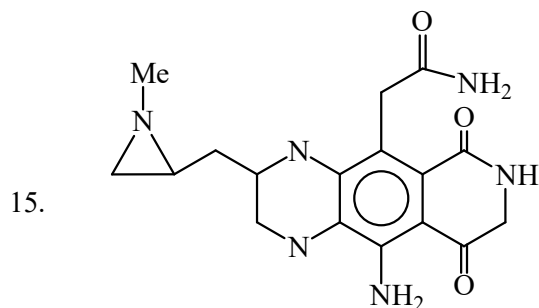
11. A quantity of 15.8 g of KMnO_4 can be decolourized in acidic medium by ($\text{K} = 39, \text{Mn} = 55, \text{Fe} = 56$)
 (A) 18.25 g HCl (B) 22.5 g $\text{H}_2\text{C}_2\text{O}_4$ (C) 32 g SO_2 (d) 38 g FeSO_4
12. Consider two reactions: (X represents an element)
Reaction-I : $\text{H}_2\text{XO}_4 + \text{KOH} \longrightarrow \text{K}_2\text{XO}_4 + \text{H}_2\text{O}$
Reaction-II : $\text{H}_2\text{XO}_4 + \text{KI} \longrightarrow \text{K}_2\text{X} + \text{I}_2$
 Which of the following statement(s) is/are incorrect?
 (A) Molar mass of H_2XO_4 is equal in both reactions.
 (B) Equivalent mass of H_2XO_4 is equal in both reactions.
 (C) Equivalent mass of H_2XO_4 in reaction-I is twice as in reaction-II.
 (D) One mole of H_2XO_4 will contain same number of equivalent in both reactions.
13. Correct IUPAC names as per rules are:
 (A) 1,1,1-trichloro-2,2-diphenylethane
 (B) 3-cyclopentylbut-1-ene
 (C) Cyclopenta-2,4-diene
 (D) 1-bromo-2,3-dichlorocyclopropane

SECTION - 2
[NON-NEGATIVE INTEGER TYPE].



Calculate the value of 'z' where

$$z = 3 \times \frac{\text{No. of } \sigma\text{-bonds}}{\text{DBE}} \times \frac{3^\circ\text{C atoms}}{\text{No. of } \pi\text{ bonds}}$$



Sum of 1° , 2° , 3° amines and amide in the above given figure is _____.

* * * * *

Answer Key**Grade : X-B****ADVANCE TEST-4****Date : 24-08 -2026****Mathematics**

1	2	3	4	5
C,D	AC	ABCD	4	25

Physics

6	7	8	9	10
BCD	ABC	ACD	262	2

Chemistry

11	12	13	14	15
AB	BCD	ABD	90	7

